

NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

FRONT END WEB DEVELOPMENT USING REACT

PROJECT REPORT

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TITLE	COVID HEALTH INTELLIGENCE DASHBOARD
GITHUB LINK	https://github.com/Jovitha05/API-Based-COVID-Health-Stats-Tracker.git

ABSTRACT

The Global Health Intelligence Dashboard is a web-based application developed using React.js to provide users with real-time COVID-19 and health-related statistics through an interactive and user-friendly interface. During global health emergencies, timely access to accurate information is essential for public awareness, decision-making, and preventive action. This project aims to address this need by collecting live health data from public APIs and presenting it in a visually appealing dashboard that allows users to monitor and analyze health statistics from different countries around the world.

The application integrates with the Disease.sh API to fetch real-time data related to total cases, recoveries, deaths, active cases, and population statistics. Users can search for specific countries and instantly view detailed health information along with the country's flag and demographic data. To enhance understanding of the collected data, the dashboard includes interactive charts and graphical visualizations that display the top affected countries and historical trends over the last thirty days. These visual representations help users easily identify patterns, compare statistics, and understand the overall health situation.

In addition to displaying numerical data, the system incorporates a risk assessment feature that categorizes countries into different risk levels based on active case counts. The dashboard also provides AI-generated health insights that summarize the current situation and help users interpret the data more effectively. A dark mode option has been included to improve accessibility and user experience, making the application suitable for different viewing environments.

The project demonstrates the practical implementation of modern frontend development concepts such as component-based architecture, state management, API integration, event handling, conditional rendering, and responsive user interface design. React Hooks such as `useState` and `useEffect` are utilized to manage application data and dynamically update the interface whenever new information is retrieved from the API.

Overall, the Global Health Intelligence Dashboard serves as an informative and analytical platform that transforms raw health data into meaningful insights. The project not only promotes awareness about public health conditions but also provides valuable hands-on experience in React.js development, data visualization, and real-world API integration. Through the successful completion of this project, a comprehensive understanding of frontend web development practices and modern application design principles was achieved.

PROBLEM STATEMENT

In today's digital world, access to accurate and real-time health information is extremely important, especially during disease outbreaks and public health emergencies. During the COVID-19 pandemic, millions of people relied on health statistics such as confirmed cases, recoveries, active cases, and death counts to understand the severity of the situation and make informed decisions. However, obtaining updated and reliable information from multiple sources can be time-consuming and confusing for users.

Many existing platforms either provide limited information or present data in a complex manner that is difficult for ordinary users to understand. In addition, some websites lack proper visualization tools, making it challenging to identify trends, compare countries, and assess the current risk level. Health awareness organizations and the general public require a centralized platform that can collect real-time data from trusted sources and display it in an interactive, user-friendly format.

Another major challenge is the absence of analytical insights that help users interpret raw statistical data. Numerical values alone may not clearly indicate whether a country is facing a high-risk or low-risk situation. Users often need graphical representations, trend analysis, and meaningful insights to better understand the health status of a region.

To address these challenges, the Global Health Intelligence Dashboard was developed. The application integrates real-time health data from public APIs and presents it through interactive charts, tables, and visual dashboards. It enables users to search for country-specific information, analyze historical trends, identify risk levels, and gain valuable insights through AI-generated recommendations. By transforming complex health statistics into an accessible and visually appealing format, the system helps improve public awareness and supports informed decision-making regarding health-related issues.

OBJECTIVES

The primary objectives of this project are:

- To develop a modern web application using React.js.
- To fetch real-time COVID-19 and health-related statistics from a public API.
- To provide accurate and up-to-date information to users.
- To design an interactive dashboard for data visualization.
- To enable users to search for country-specific health information.
- To display statistics such as total cases, active cases, recoveries, and deaths.
- To implement graphical representations using charts and graphs.
- To perform trend analysis using historical health data.
- To identify and categorize countries based on risk levels.
- To generate AI-based health insights for better understanding of data.
- To implement dark mode functionality for enhanced user experience.
- To improve awareness regarding global health situations through technology.

PROJECT SCOPE

The Global Health Intelligence Dashboard focuses on providing users with real-time access to global and country-specific health statistics. The application serves as an informative platform where users can view live health data, analyze trends, and gain insights regarding health conditions across different regions of the world.

The project includes features such as global statistics monitoring, country-wise search functionality, graphical visualization of data, risk-level assessment, and trend analysis. Through the integration of public APIs, the system ensures that users receive updated and accurate information without manual intervention.

The dashboard is designed with a user-friendly interface that enables users from different backgrounds to easily access and understand health-related information. Interactive charts and tables make data interpretation easier and more meaningful.

Current Scope:

- Real-time health statistics retrieval
- Country-specific search functionality
- Interactive charts and visualizations
- Historical trend analysis & Risk-level categorization
- AI-generated health insights
- Responsive user interface and Dark mode functionality

Future Scope:

- Multi-disease monitoring
- Predictive analytics using machine learning
- User account management & Mobile application support
- Exporting reports in PDF format
- Integration with healthcare organizations

TECHNOLOGIES USED

The project was developed using modern web development technologies and tools to ensure performance, scalability, and maintainability.

Frontend Technologies:

- React.js
- HTML5
- JavaScript (ES6)
- CSS3

Libraries and Frameworks:

- Axios
- React ChartJS 2
- Chart.js
- React Hooks

Development Tools:

- Visual Studio Code
- Git
- GitHub

API:

- Disease.sh API

Browser:

- Google Chrome

Operating System:

- Windows

These technologies collectively contributed to building an interactive, responsive, and efficient web application.

APPLICATION FLOW

The application follows a systematic workflow to provide users with accurate and real-time health information.

Step 1:

The user opens the application through a web browser.

Step 2:

The dashboard automatically fetches global COVID-19 statistics from the Disease.sh API.

Step 3:

The system displays worldwide statistics including total cases, recoveries, deaths, and active cases.

Step 4:

The application retrieves data for all countries and generates visual charts showing the top affected countries.

Step 5:

Users enter a country name in the search box.

Step 6:

The system sends a request to the API and retrieves country-specific information.

Step 7:

Detailed statistics such as population, total cases, active cases, recoveries, and deaths are displayed.

Step 8:

The application analyzes active cases and determines the corresponding risk level.

Step 9:

Historical data for the selected country is fetched and displayed using trend charts.

Step 10:

AI-generated health insights are provided to help users better understand the country's current health status.

Step 11:

Users can switch between light mode and dark mode based on their preferences.

KEY REACT CONCEPTS USED

1. Components

The application is divided into reusable components such as DashboardCards, SearchCountry, TrendChart, RankingTable, AIInsight, and DarkModeToggle. This improves code organization and maintainability.

2. JSX

JSX allows HTML-like syntax within JavaScript, making UI development easier and more readable.

3. useState Hook

The useState hook is used to manage dynamic data such as health statistics, country details, trend data, and dark mode status.

4. useEffect Hook

The useEffect hook is utilized to fetch API data when the component loads and update the application dynamically.

5. Props

Props are used to transfer data between parent and child components, enabling effective communication within the application.

6. Event Handling

Event handling is implemented for user actions such as searching countries and toggling dark mode.

7. Conditional Rendering

Conditional rendering ensures that charts, country details, and AI insights are displayed only when relevant data is available.

8. API Integration

Axios is used to connect with external APIs and retrieve real-time information efficiently.

9. State Management

State management helps maintain application data consistency and supports dynamic updates.

10. Responsive Design

React components are designed to adapt to different screen sizes, providing a consistent user experience across devices.

OUTPUT

The application successfully generates a complete health analytics dashboard that provides real-time information to users.

Expected outputs include:

- Display of global health statistics
- Country-specific data retrieval
- Interactive bar charts
- Historical trend analysis charts
- Risk-level identification
- AI-generated health insights
- Responsive dashboard interface
- Dark mode functionality

Screenshots:

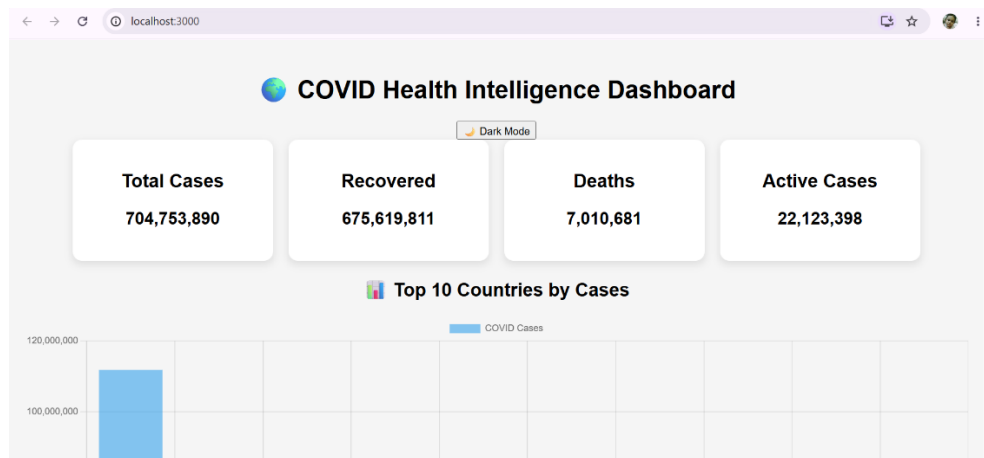


FIG.1 GLOBAL STATISTICS DASHBOARD

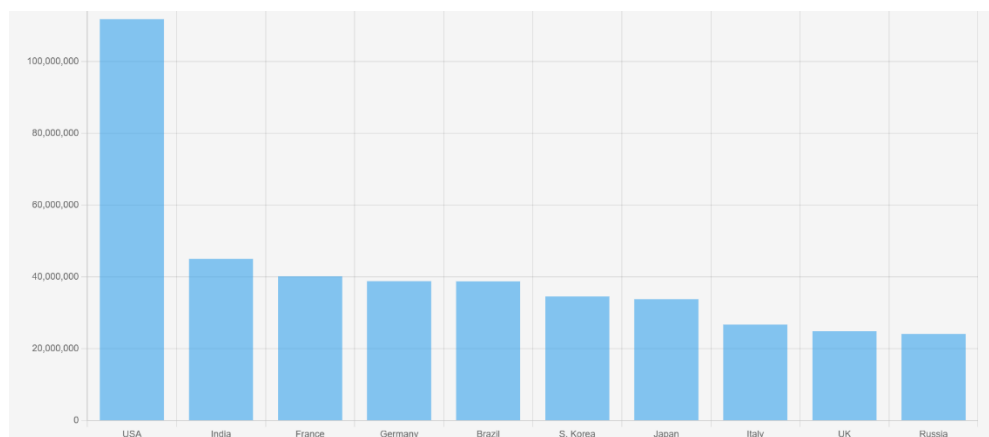


FIG.2 TOP 10 COUNTRIES CHART

🏆 Top 10 Countries Ranking

Rank	Country	Cases
1	USA	111,820,082
2	India	45,035,393
3	France	40,138,560
4	Germany	38,828,995
5	Brazil	38,743,918
6	S. Korea	34,571,873
7	Japan	33,803,572
8	Italy	26,723,249
9	UK	24,910,387
10	Russia	24,124,215

FIG.3 RANKING TABLE OF TOP 10

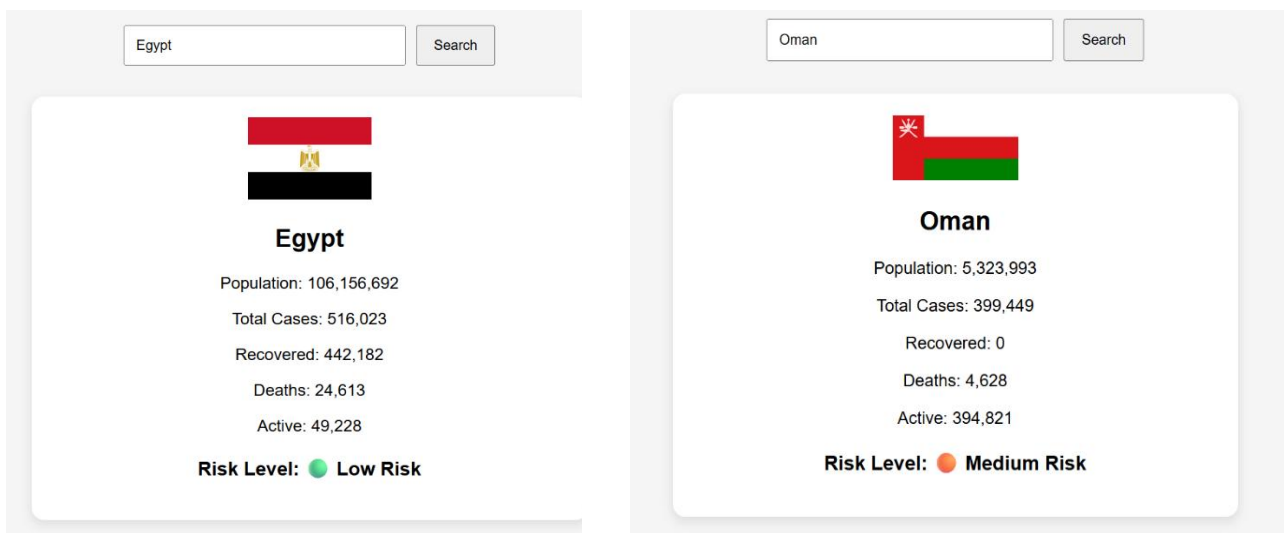


FIG.4 RISK ASSESSMENT DISPLAY

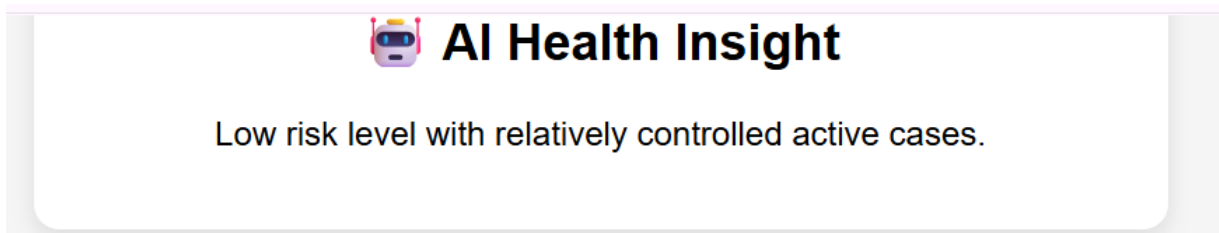


FIG.5 AI HEALTH INSIGHTS

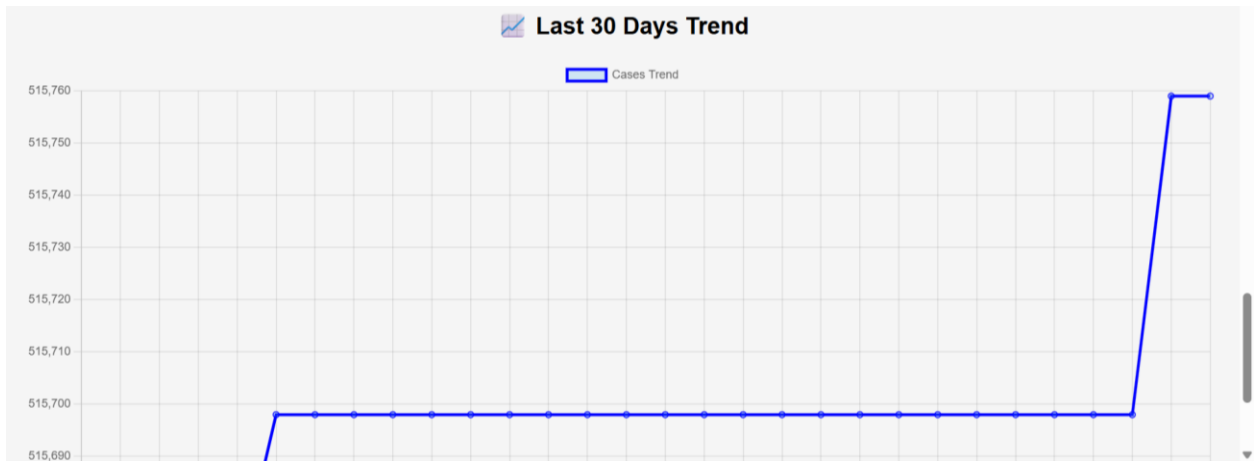


FIG.6 HISTORICAL TREND ANALYSIS



FIG.7: DARK MODE INTERFACE

CHALLENGES FACED

Several challenges were encountered during project development.

- Understanding the React component structure.
- Managing multiple state variables efficiently.
- Integrating external APIs and handling asynchronous operations.
- Debugging API response issues.
- Designing responsive and visually appealing interfaces.
- Implementing chart visualizations using Chart.js.
- Handling invalid user inputs during country searches.

Solutions Implemented:

- Extensive practice with React Hooks.
- Proper error handling using try-catch and promise-based methods.
- Testing API endpoints before integration.
- Continuous debugging and UI improvements.

These challenges provided valuable practical learning experiences.

LEARNING OUTCOMES

Through the successful completion of this project, several technical and professional skills were acquired.

Technical Skills:

- React.js Development
- API Integration
- State Management
- Component-Based Architecture
- Data Visualization
- Responsive Design
- JavaScript Programming

Professional Skills:

- Problem Solving
- Debugging
- Project Planning
- Documentation
- Version Control using GitHub
- Analytical Thinking

The project provided hands-on experience in developing real-world web applications using modern technologies.

CONCLUSION

The Global Health Intelligence Dashboard was successfully designed and implemented using React.js and modern web development practices. The application effectively retrieves and displays real-time health statistics through a clean, interactive, and user-friendly interface. Features such as country-wise search, graphical data visualization, risk-level assessment, trend analysis, and AI-generated health insights enhance the usefulness of the platform and improve user understanding of health-related information.

The project demonstrates practical knowledge of React concepts, API integration, state management, and frontend application development. It successfully addresses the problem of accessing and interpreting real-time health data by transforming complex statistics into meaningful insights. Overall, the project achieved its objectives and provided valuable experience in building scalable and interactive web applications. Future enhancements can further extend its capabilities and usefulness in public health monitoring and awareness.